

Purnendu Tiwari

Rewa, India | +91 9685021295 | purnendutiwari2004@gmail.com
linkedin.com/in/purnendutiwari | github.com/purnendu5804

Experience

Software Development Intern

May 2026 – July 2026

FOSSEE, IIT Bombay

Mumbai, India (Remote)

- Collaborated with a mentor and a 3-member team to modernize the **Linux packaging pipeline** for the eSim application across **AppImage** and **Flatpak** formats
- Fixed **PyQt5** dependency crashes by bundling a self-contained **Python runtime** into the AppImage, removing all host-side setup requirements
- Resolved **Qt platform plugin** failures by building automatic **X11/Wayland** detection logic, fixing rendering across Linux desktop environments
- Rebuilt the **Flatpak** offline install pipeline by fixing broken package sources and pre-validating dependency wheels, enabling zero-network builds

Projects

sdcHouse – Real-time Virtual Space | *Next.js, WebRTC, Redis*

Live | [GitHub](#)

- Developed a **metaverse-lite** virtual space delivering real-time **spatial audio** and video communication for **10+ concurrent users**
- Designed a **queue-based pub/sub architecture** that cut backend latency by **25%**, keeping collaboration smooth at scale

RCE Engine – Remote Code Execution Platform | *TypeScript, BullMQ, Docker, Prisma*

[GitHub](#)

- Architected a remote code execution engine supporting **multi-language compilation** and execution inside isolated, **sandboxed containers**
- Designed an **asynchronous job pipeline** to queue and process concurrent code submissions with horizontally scalable workers
- Implemented **RESTful APIs** and backend services with **Prisma** to manage submissions and container lifecycles reliably

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, SQL, Bash

Web Development: React JS, Node JS, Express JS, REST APIs, MongoDB, MySQL, PostgreSQL, Next.js, Prisma

Cloud & DevOps: AWS (S3, CloudFront, IAM, VPC, EC2), Docker, Terraform, GitHub Actions, CI/CD, Prometheus, Grafana, Cloudflare, Linux

Developer Tools: Git, GitHub, Bitbucket, Jira, Redis, BullMQ, Postman

Core Concepts: Data Structures & Algorithms, System Design, Distributed Systems, OOP, Agile, Scrum, SDLC

Education

Vellore Institute of Technology (VIT)

Bhopal, India

B.Tech in Computer Science; CGPA: 8.97/10.0

2023 – 2027

Relevant Coursework: Data Structures & Algorithms, OS, DBMS, OOP, Computer Networks

Certifications

AWS Certified Solutions Architect – Associate (SAA-C03) – Amazon Web Services

Introduction To Machine Learning – NPTEL (IIT Madras)

Achievements & Co-curricular

Leadership: President, Software Development Club, VIT Bhopal; led a 6-member team to organize 5+ technical workshops, hackathons, and coding events for 300+ students

LeetCode Knight: Achieved Highest Contest Rating **1852** and Among the Top 5% of coders on LeetCode

GirlScript Summer of Code: Ranked **419th** among global open-source contributors

Youth Parliament: 1st Runner Up, Zonal-Level Competition

National Debate: 2nd Runner Up, Ek Bharat Shreshth Bharat Competition

Music: Diploma in Classical Music (Raja Mansingh Tomar Music Arts University)